LIMITS OF REDUCED INVERSE SYSTEMS ARE REDUCED

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1009-1021, following the statement at 1001-1007.

Editorial note. No omissions; the filtered-colimit-of-reduced-rings observation is displayed in the source.

Allowed standard prerequisites (editorial).

- Existence of directed inverse limits. lemma-directed-inverse-system-has-limit (same chapter, card L26291, cited at ~1005-1006). lim X_i exists.

Source reference. Original lemma 07SF: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 878–878. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let I be a directed set. Let $(X_i, f_{i'i})$ be an inverse systems over I of algebraic spaces over S. If X_i is reduced for all i, then X is reduced.*

Proof. Observe that $\lim X_i$ exists by Lemma 07SF. Pick $0 \in I$ and choose an affine scheme V_0 and an étale morphism $U_0 \to X_0$. Then the affine schemes $U_i = X_i \times_{X_0} U_0$ are reduced. Hence $U = X \times_{X_0} U_0$ is a reduced affine scheme as a limit of reduced affine schemes: a filtered colimit of reduced rings is reduced. Since the étale morphisms $U \to X$ form an étale covering of X as we vary our choice of $U_0 \to X_0$ we see that the lemma is true. □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.